Homework #8

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Contents

Introduction

I'm Advay Vyas, EID: av37899, and this is my submission for SDS 315 Statistical Thinking Homework #8. The GitHub repository for my code is at this link.

Problem 1

Part A

$$\hat{y} = 147.81 - 0.62x$$

After fitting the linear regression model (above), we ended with a y-intercept of 147.81 and a slope of -0.62. Therefore, to calculate the clearance rate for a 55-year old, we can just do the manual calculation of the intercept plus 55 times the slope. That results in a creatinine clearance rate of 113.72.

Part B

Creatinine clearance rate changes with age at a rate of -0.62 mL/minute per year. This value was determined through a linear regression model and this value is the slope of that model.

Part C

To find whose clearance rate was higher for their age, we calculated the residual for each of these individuals. The predicted value for the 40-year-old was calculated with the intercept + 40 times the slope to result in 123.02. Similarly the 60-year-old had a predicted value of intercept + 60 times the slope for 110.62.

Next, we just subtracted their predicted values from their actual values since the actual values are greater and figured out that since 11.98 is greater than 1.38, the 40-year-old has a higher rate for his age according to the linear regression.

Problem 2

Part A

To calculate these values, we first bootstrap COVID data (filtered by country equaling Italy) 10,000 times, fit a linear regression model each time, and then create a data set of the coefficients. From that data set, we get the estimate and use mosaic's confint function to calculate confidence intervals.

Estimate for Growth Rate in Italy: 0.18

95% Confidence Interval for Growth Rate in Italy: [0.1591137, 0.2079875]

For the doubling time, we divide 70 by the growth rate which follows from the established rule of 70 in class. A similar process to growth rate is followed to generate the values below.

Estimate for Doubling Time in Italy: 3.83

95% Confidence Interval for Doubling Time in Italy: [3.365587, 4.3993691]

Part B

Like the Italy version, these values are created by bootstrapping and in-built R functions.

Estimate for Growth Rate in Spain: 0.28

95% Confidence Interval for Growth Rate in Spain: [0.2343474, 0.3180995]

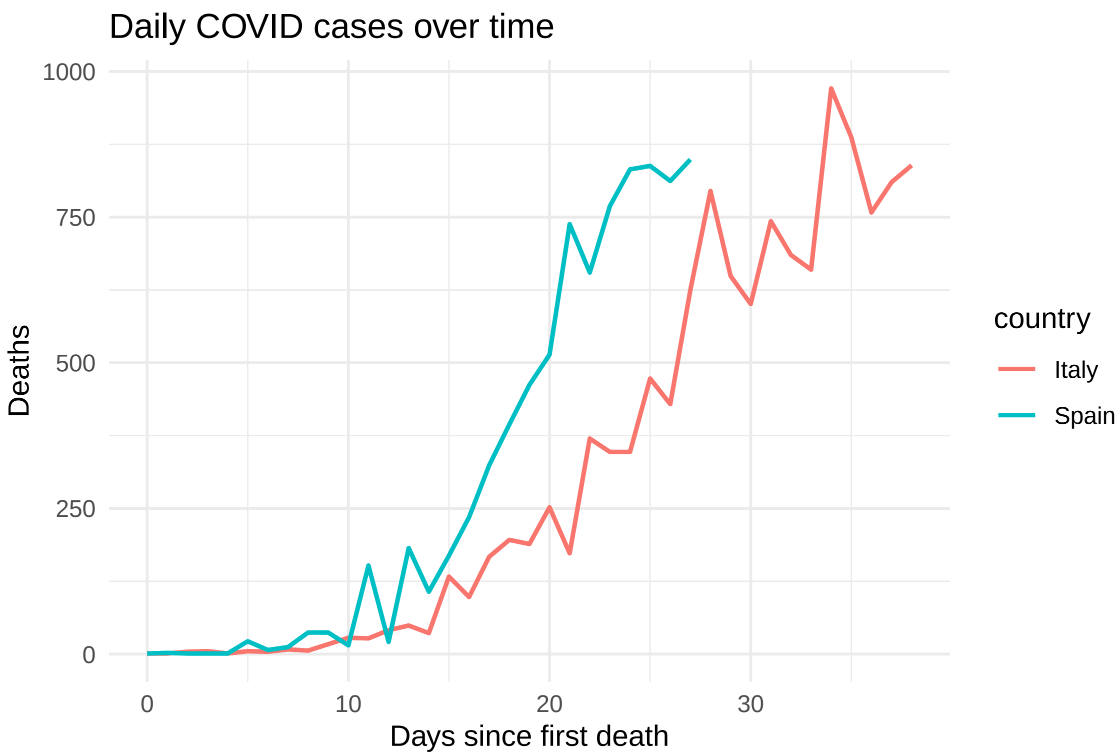
For the doubling time, a similar process follows and creates the below values.

Estimate for Doubling Time in Spain: 2.55

95% Confidence Interval for Doubling Time in Spain: [2.2005693, 2.9870178]

Part C

This line graph shows daily COVID cases over time, split into a line for Italy and a line for Spain.



Problem 3

The formula for a power-law relationship is $y = Kx^{\beta}$, where K is the initial value of y and β is the elasticity (proportional change in y with respect to x). To relate this idea to our example, we attempt to fit a power-law relationship to our data.

From the textbook, we know we can break up that earlier equation into $\log y = \log K + \beta \log x$. Therefore, if we can fit a linear regression model that relates $\log y$ and $\log x$, then the slope of that regression model will be β.

Now, let y represent sales and x represent price, which implies that β is the price elasticity of milk by the definition above. Fitting the aforementioned linear regression model, we receive the estimate and confidence interval below.

Estimate for Price Elasticity of Milk: -1.62

95% Confidence Interval for Price Elasticity of Milk: [-1.7726551, -1.4532995]